

# RC Design Project 01

We can delete this since it no need



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# 1 How to Design

- Building Load contains : **gravity** and **lateral** Loads.

## 1.1 Convert Area Loads to Line Load on Beams

Slabs distribute their load to beam depending on slab span direction and support condition (EN 1992-1-1:2023, Cl. 5.3.2 [1]; IStructE Manual, Cl. 3.2).

**The Tributary Width Method [IStructE Manual, Cl. 3.2]:**

For a **one-way slab**, load goes to the beams on the short sides:

$$\text{Beam Line Load (kN/m)} = \text{Floor Load (kN/m}^2\text{)} \times \text{Tributary Width (m)}$$

**Tributary Width** = half the slab span on each side of the beam [IStructE Manual, Cl. 3.2].

**Example:** If Slab spans 5m between two beams, so each beam take  $5/2 = 2.5m$  tributary width. And if floor load is  $10\text{kN/m}^2$ , then beam line load is  $10 \times 2.5 = 25\text{kN/m}$

For a **two-way slab**, load distributes in both directions using yield line theory or the simplified triangular/trapezoidal method (EN 1992-1-1:2023, Cl. 5.3.2 [1]; Reynolds & Steedman, Ch. 8):

- Short span beam take triangular load
- Long span beam take trapezoidal load

## 1.2 Apply ULS and SLS Load Combinations

- ULS (Ultimate Limit State) - for strength design

$$\sum_i \gamma_{G,i} G_{k,i} + \gamma_{Q,1} Q_{k,1} + \sum_{j>1} \gamma_{Q,j} \varphi_{0,j} Q_{k,j}$$

- SLS (serviceability Limit State) - for deflection/crack check

Characteristic Combinations:

$$\sum_i G_{k,i} + Q_{k,1} + \sum_{j>1} \varphi_{0,j} Q_{k,j} + (P_k)$$

Quasi-permanent (for long-term deflection)

$$\sum_i G_{k,i} + \sum_j \varphi_{2,j} Q_{k,j} + (P_k)$$



1.3 Beam Support Conditions

The support condition effects bending moments, shear forces, and reinforcement layout (EN 1992-1-1:2023, Cl. 5.3.2.2 [1]; IStructE Manual, Cl. 4.2):

Table 1.1: Beam support condition

| Support Type     | Description                   | Moment at Support          |
|------------------|-------------------------------|----------------------------|
| Simply supported | Beam sits on wall/pad         | Zero moment at ends        |
| Continuous beam  | Multiple spans, monolithic    | Hogging moment at supports |
| Fixed end        | Fully restrained (rare in RC) | Full fixity moment         |
| Partially fixed  | Column/beam junction          | Partial moment transfer    |

| Support Type     | Rotation Allowed | Moment at Support | Typical Structure     |
|------------------|------------------|-------------------|-----------------------|
| Simply Supported | Yes              | 0                 | Beam on columns       |
| Continuous       | Small rotation   | Negative moment   | Multi-span floor beam |
| Fixed            | No rotation      | Maximum moment    | Cantilever            |
| Partially Fixed  | Limited rotation | Partial moment    | Beam-column joint     |

In real RC buildings: Most beams are continuous over columns, this is the most common and important case (EN 1992-1-1:2023, Cl. 5.3.2 [1]; IStructE Manual, Cl. 4.2).

## 2 Introduction

This is Our Team Intro.

### 3 Detailed Design Calculations

#### 3.1 Loading

##### 3.1.1 Live Load

For imposed loads on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]. Is summary in Table 3.1.

Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category   | Sub-category / Use                                                                  | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|------------|-------------------------------------------------------------------------------------|-------------------------------|---------------------|
| Category A | Areas for domestic and residential activities                                       |                               |                     |
|            | - Floors                                                                            | 1,5 to 2,0                    | 2,0 to 3,0          |
|            | - Stairs                                                                            | 2,0 to 4,0                    | 2,0 to 4,0          |
|            | - Balconies                                                                         | 2,5 to 4,0                    | 2,0 to 3,0          |
| Category B | Office areas                                                                        | 2,0 to 3,0                    | 1,5 to 4,5          |
| Category C | Areas where people may congregate                                                   |                               |                     |
| C1         | Areas with tables<br>(e.g. schools, cafes, restaurants, reading rooms)              | 2,0 to 3,0                    | 3,0 to 4,0          |
| C2         | Areas with fixed seats<br>(e.g. theatres, cinemas, churches)                        | 3,0 to 4,0                    | 2,5 to 7,0<br>(4,0) |
| C3         | Areas without obstacles for moving people<br>(e.g. museums, entrance halls)         | 3,0 to 5,0                    | 4,0 to 7,0          |
| C4         | Areas with possible physical activities<br>(e.g. dance halls, gyms)                 | 3,5 to 5,0                    | 3,5 to 7,0          |
| C5         | Areas susceptible to large crowds<br>(e.g. concert halls, sports halls with stands) | 5,0 to 7,5                    | 3,5 to 4,5          |
| Category D | Shopping areas                                                                      |                               |                     |

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Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category          | Sub-category / Use                                                       | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|-------------------|--------------------------------------------------------------------------|-------------------------------|---------------------|
| D1                | Areas in general retail shops                                            | 4,0 to 5,0                    | 3,5 to 7,0<br>(4,0) |
| D2                | Areas in department stores                                               | 4,0 to 5,0                    | 3,5 to 7,0          |
| <b>Category E</b> | Storage and industrial activities<br>(general storage, accessible areas) | 7,5                           | 7,0                 |

And this is the imposed loads on second floor that we assumed. which shown in Table 3.2.

Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                      | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|----------------------------------|---------------------------|-------------------------------|
| 1   | Receptionist/<br>Reception       | B                         | 3                             |
| 2   | Workstation /<br>Open office     | B                         | 3                             |
| 3   | Meeting Room                     | C1                        | 3                             |
| 4   | President Office                 | B                         | 3                             |
| 5   | Vice President Offices           | B                         | 3                             |
| 6   | Accounting                       | B                         | 3                             |
| 7   | General Sale &<br>Administration | B                         | 3                             |
| 8   | Copy /<br>Printer Area           | B                         | 3                             |
| 9   | Storages                         | E1                        | 5                             |
| 10  | Lift Lobby /<br>Lift Area        | C3                        | 4                             |
| 11  | Stair (Main Stair)               | A                         | 4                             |
| 12  | Restroom (Men)                   | B                         | 3                             |
| 13  | Restroom (Women)                 | B                         | 3                             |
| 14  | —                                | —                         | —                             |
| 15  | —                                | —                         | —                             |
| 16  | —                                | —                         | —                             |

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Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|----------------------------|---------------------------|-------------------------------|
| 17  | Entrance                   | C3                        | 4                             |
| 18  | Emergency Stair            | A                         | 4                             |
| 19  | Server &<br>Technical Room | E1<br>(conservative)      | 5                             |
| 20  | Kitchen /<br>Pantry        | C1                        | 3                             |
| 21  | Library                    | C3                        | 4                             |
| 22  | Share Space                | C3                        | 4                             |

### 3.1.2 Dead Load

According to EN 1991-1-1:2025, Table A.1 [2]; EN 1991-1-1:2002, Cl. 4 [3]:

Table 3.3: Dead Load Typical Value

| Source                                 | Typical Value                          |
|----------------------------------------|----------------------------------------|
| Reinforced concrete slab (200mm thick) | $0.2 \times 25 = 5.0 \text{ kN/m}^2$   |
| Screed/floor finish                    | $1.5 \text{ kN/m}^2$                   |
| Ceiling + services (MEP)               | $0.5 \text{ kN/m}^2$                   |
| Partitions (treated as permanent)      | $1.0 \text{ kN/m}^2$                   |
| <b>Total <math>G_k</math></b>          | <b><math>8.0 \text{ kN/m}^2</math></b> |

## References

- [1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.
- [2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.
- [3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.
- [4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.
- [5] EN 1990, *Basis of Structural Design*
- [6] IStructE Manual, *Manual for the Design of Concrete Building Structures to Eurocode 2*
- [7] Reynolds & Steedman, *Reinforced Concrete Designer's Handbook, 10th Ed.*

*Note that :* We may use the referennces provide by professor (check in telegram group) if it still not enough we can add more maybe.

## Appendices